

Pedro Librado Uribe Reyes

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Engineering student building edge AI, embedded, and robotics systems, from 4-bit quantized models designed for custom silicon to LiDAR and YOLO perception on an autonomous boat. Hackathon winner, world-level robotics champion, and founder of an AI product that has generated about US\$1,500.

EDUCATION

Tecnológico de Monterrey — *B.S. in Robotics and Digital Systems Engineering, Monterrey, Mexico*

5th semester, in progress

CETYS Preparatoria — *International Baccalaureate, GPA 9.5/10*

TECHNICAL SKILLS

Languages: Python, C/C++, Java, TypeScript, Verilog, VHDL. Spoken: Spanish (native), English (fluent)

Edge AI and vision: TensorRT, ONNX, YOLO, OpenCV, scikit-learn, model quantization, NVIDIA Jetson

Embedded and hardware: STM32, ESP32, FPGA (Intel MAX 10 on DE10-Lite, Xilinx ZYNQ), RTOS, UART, CAN

Robotics: ROS 2, Gazebo, LiDAR, PID and feedforward control, WPILib

Tools: Linux, Docker, Git, PostgreSQL, Redis, FastAPI, Flask

EXPERIENCE

USV Perception Lead, VantTec (Tecnológico de Monterrey)

2025 – Present

- Lead perception for an autonomous surface vehicle at RoboBoat 2026; rebuilt the ROS 2 vision pipeline after its stereo camera failed, fusing a Velodyne VLP-16 LiDAR with a YOLO detector to publish 3D object positions.
- Wrote the LiDAR-to-bounding-box ranging node (median range with MAD outlier rejection): ~6.2 m against ~7 m ground truth in simulation, and markers measured out to ~18 m on recorded boat data.
- Trained the YOLO buoy dataset, compiled the TensorRT engine, and validated in Gazebo Harmonic before water tests; built SocketCAN, Xbee telemetry, and remote-terminal links for field testing.

Research Assistant, University of California San Diego (VLSI Information Processing Lab)

Jun – Aug 2023

- Designed a real-time shape classifier (linear SVM) with weights quantized to 4-bit integers for hardware implementation, holding 98%+ accuracy on 3,000 test images; worked in Verilog on a Xilinx ZYNQ board.
- Presented at the UCSD closing conference and as an invited speaker at the UCSD-ENLACE-México Symposium (Instituto Tecnológico de Tijuana).

Software Lead and Captain, Innobotics FRC Team 6694

Feb 2022 – Jan 2025

- Owned robot code in Java/WPILib over three seasons: subsystems, autonomous routines, PID and feedforward control tuned from SysId measurements, and NetworkTables telemetry.
- Led the team to 3rd place and the Spirit Award at Regional Hermosillo (2024) and 5th place with the Imagery Award at Regional Monterrey (2023).

Software Lead and Captain, Mexicali Robotics

2016 – 2022

- Six years of competitive robotics; 3rd place worldwide at Infomatrix World Finals, Bucharest (2019).

PROJECTS

SentinelBox, edge-AI HVAC predictive maintenance *1st place, Carrier Keep It Cool Hackathon 2026*

github.com/ayexxxd/SentinelBox

- Designed the electronics and UART link between sensor nodes and an ESP32 gateway that learns each unit's baseline and runs a compact neural network on-device, with no cloud dependency.
- Controlled prototype validation: 6 s average detection time and 0% false alarms after a 90 s baseline learning period.

Uribot, AI automation platform for small businesses *Founder*

2025 – Present | uribot.chat

- Built a multi-tenant system (Python, Flask, PostgreSQL, Redis, Docker) that turns WhatsApp, Instagram, and Messenger chats into quotes and bookings with human handoff; about US\$1,500 in revenue to date.

MiNorte, AI financial analyst for small businesses *HackMTY 2026, Banorte Challenge*

github.com/pedroolib/MiNorte-by-Banorte

- Built the deterministic financial engine, FastAPI backend, and MCP tools that let AI agents explain verified numbers and generate dashboard cards.

Smart Tractor, embedded vision

github.com/pedroolib/JohnDeere-RLTractorModelation

- STM32 tractor written in bare-metal C and RTOS, controlled with Logitech pedals, with a Grafana dashboard and vision-based operator drowsiness detection.

FPGA Controller, custom hardware controller *Terasic DE10-Lite*

- Built a custom hardware controller in VHDL on a DE10-Lite FPGA that drives a simulator through real-time UART communication.

HONORS AND LEADERSHIP

- 1st place worldwide, NASEF Minecraft Masters (Dec 2020) and NASEF Farmcraft (Jul 2020), representing Mexico
- 3rd place worldwide, Lego Sumo, Infomatrix World Finals (2019); 2nd place continental, Robomatrix (2018)
- FTC robotics mentor, Botbusters (2025); robotics education volunteer, SMALC (2026)